Dual-Path Prototype Feature Decoupling Alignment Network for Panchromatic and Multispectral Classification

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***Abstract*— In recent years, with the rapid advancements and widespread application of satellite photography technology, it has become increasingly possible to obtain high-quality panchromatic (PAN) and multispectral (MS) data, which has provided new opportunities and challenges for multisource information fusion and classification research. Remote sensing data have the characteristics of small interclass differences and large intraclass differences, which easily leads to category confusion in network learning. In addition, how to fully tap the advantages of multisource data, better align multisource features, improve classification accuracy, and achieve collaborative classification are key issues that need to be solved urgently. In this article, a dual-path prototype feature decoupling alignment network (DPFDA-Net) is designed to solve the above issues. The network consists of two components: a prototype feature embedding (PFE) module and a feature alignment module (FAM) based on prototype decoupling. In the feature extraction stage, the PFE module uses the prototype concept to learn the discriminative prototype features of each category of the dual-source data separately, making the boundaries between categories more obvious. The FAM operates at the dual-source prototype feature level and achieves feature alignment by decoupling single-source prototype features and performing feature transformation to supplement the missing information of another data source. Finally, we use the aligned features for classification. The results of the experiment demonstrate that our approach has made significant progress in improving classification precision. The code is available at https://github.com/Xidian-AIGroup190726/DPFDANet.**

***Index Terms*— Adaptive instance normalization (AdaIN), deep learning, dual-source RS data, feature alignment, feature decoupling, prototype learning.**

I. Introduction

IN RECENT years, great progress has been made in the fields of Earth observation, land use, and environmental monitoring, owing to the swift advancement of satellite technology. Remote sensing images captured by satellites provide a wealth of ground object information [1], [2]. However, due to differences in spatial and spectral resolutions between panchromatic (PAN) images and multispectral (MS) images, using only one type of data for ground object classification often fails to meet accuracy requirements [3], [4]. PAN images have the advantage of high spatial resolution and can capture detailed features of ground objects. However, due to the lack of spectral information, its ability to identify spectral features is weak. In contrast, MS images provide rich spectral information and can describe the spectral characteristics of ground objects, but their spatial resolution is relatively low [5], [6], [7], [8]. To solve the above problems, current research mainly uses solutions based on dual-source data fusion to give full play to their respective advantages to improve the ability of ground object classification [9], [10], [11].

The scheme based on dual-source data fusion classification overcomes the technical limitations of obtaining high-resolution remote sensing images for object recognition and has received extensive research. Currently, there are two types of fusion classification schemes based on dual-source data: 1) data-level fusion classification and 2) feature-level fusion classification. Whether it is the former or the latter, they aim to make full use of the advantages of dual-source data and improve the precision and reliability of ground object classification through fusion. The development of these methods provides strong technical support for remote sensing image recognition.

For data-level fusion classification methods, the general process is to first decompose dual-source data, fuse their dominant components to obtain a high-resolution fusion image, and then extract features from the fused image for classification. There are two main strategies for data-level fusion: component substitution (CS) [12], [13], [14] and multiresolution analysis (MRA) [15], [16], [17]. CS is based on the assumption that data acquired by different sensors are complementary in some aspects, such as PAN and MS. This method enhances data information by selecting components of one sensor (e.g., PAN images) to replace the corresponding components of another sensor (e.g., MS images). In [18], an adaptive IHS strategy is proposed to align various dimensions of different sensor data, and a mask is designed to reduce the difference between dual-source data components, and then data fusion is performed to lay a good foundation for subsequent network feature extraction and fusion classification. MRA is designed to take advantage of the different resolution characteristics of different sensors. It first registers data at different resolutions.

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Fig. 1. Intra- and interclass differences. To identify the category, we overlay the MS image with the ground-truth image. (a) Bareland and (b) sparse building, and (c) sparse building and (d) vegetation are two groups of objects belonging to different categories, and their spectra and detail textures show relatively consistent characteristics. (e)–(g) Same category of objects: white building, but their sparsity, spatial details, and scale information show great differences.

These data are then fused to obtain a fused image with higher resolution and richer information. Wang et al. [19] design a dual-scale regression model combined with MRA to enrich the scale information of features. This model has good adaptability to different scenarios. Zhang et al. [20] use the wavelet transform to realize information exchange and fusion between different frequencies while focusing on learning high-frequency information and constructing a wavelet network for PAN sharpening. Although these works have achieved good fusion results, they are not conducive to the end-to-end training of classification networks.

For feature-level fusion classification methods, discriminative features are usually extracted from dual-source data separately, and then they are fused in the feature space for classification. Si et al. [21] combine the advantages of two feature learning methods, discriminative and generative, and use probability distributions to model the prediction of integrated features to generate reliable classification results. Ma et al. [22] used an improved SA mechanism to enhance single-source information and then fuse dual-source features, effectively alleviating modal differences. At the same time, the loss function is designed to promote complementary learning of the dual branches. Zhao et al. [23] design a transformer network based on Fourier transform to extract local and global features. The network is faster than the general transformer architecture and can extract rich context. However, these methods focus more on discriminative feature extraction and ignore the problem of large intraclass differences and small interclass differences in remote sensing data itself, as shown in Fig. 1.

The significant progress achieved by the above research results provides us with new ideas for further exploring the construction of a dual-source remote sensing data fusion classification network. However, we also recognize some potential challenges that need to be overcome as follows.

1) Different types of ground objects in remote sensing data usually have different characteristics. However, samples of the same ground object category may have diversity, such as different shapes, textures, and grayscale features. In addition, there may be similar spectral features between different ground object categories. For example, different vegetation types may have similar spectral features. The boundaries between categories are not clear enough, making them easy to confuse during classification.
2) PAN and MS data come from different sensors, and their feature expressions are different. At the same time, the independent extraction of category features by two branches may lead to inconsistency of features in the embedding space, which brings challenges to the collaborative classification of dual-source data.

To deal with the above problems, this article designs a dual-path prototype feature decoupling alignment network (DPFDA-Net) to achieve dual-source data collaborative classification. The research work of this article is summarized as follows.

1) To address the problem of unclear category boundaries, we designed a prototype feature embedding (PFE) module. The prototype concept is used to learn discriminative category prototypes from dual sources, thereby enhancing intraclass compactness and interclass separability, and mining representative category features. At the same time, to avoid large deviations in the learned prototypes, the prototype contrast learning loss is introduced to supervise the dual-source prototype feature learning.
2) To reduce the influence of feature expression differences on classification and obtain a consistent feature embedding space, we propose a feature alignment module (FAM) based on prototype decoupling. This module decouples prototype features, supplements the missing information of the single source, and establishes information symmetry between dual sources, thereby achieving dual-source prototype-level feature alignment, which is more accurate than ordinary feature-level alignment.

The subsequent sections of this article are structured as follows. In Section II, a comprehensive review of the relevant research is presented. Section III provides a detailed description of our design methodology. Section IV describes the datasets used in this study and verifies the effectiveness of the design method. Section V offers the concluding remarks of this article.

II. Related Work

In this part, we briefly describe some techniques and modules related to our design approach, including prototype network, contrastive learning, and adaptive instance normalization. The details are as follows.

A. Prototype Network

The prototype network is a neural network model based on distance metrics, and its design is inspired by the prototype theory. This theory holds that people usually classify and

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identify objects by classifying things into prototypes or representative instances. Its core idea is to represent the differences between different categories by learning prototypes of various types of samples.

During the application process, the prototype network can be combined with CNN for end-to-end training [24]. This process can be described as follows:

$$\hat{y} = \arg\min_{c} d\left(x, \frac{1}{N_c}\sum_{i=1}^{N_c} f(x_i)\right) \tag{1}$$

where $\hat{y}$ is the prediction result. c is the category index. x is the input prediction sample. N_c is the total number of samples of category c. x_i represents the ith sample in category c. $f(\cdot)$ represents the CNN feature extractor. $d(x, y)$ refers to the distance metric function.

First, use CNN to extract the feature representation of the target; then, obtain the prototype representation of each category by averaging the feature vectors of each category; finally, calculate the distance between the input sample and the prototype vector of each category, with the minimum distance The prototype category is used as the prediction result [25].

In the domain of remote sensing data recognition, prototype networks can be applied to solve multicategory remote sensing image classification problems [26], [27], [28]. Remote sensing images are large-scale, multicategory, and high-dimensional. Traditional classification methods often face challenges such as sample imbalance [29], feature redundancy [30], and high computational complexity [31]. The prototype network can effectively reduce sample redundancy by learning the prototype vector of each category, and it also has a certain degree of robustness to the problem of sample imbalance. Cheng et al. [32] use prototype networks to integrate class imbalance and semi-supervised learning, rebalance features, and measure the similarity between samples, effectively improving the detection performance. In [33], to address the challenge of insufficient samples of remote sensing scenes, Wang et al. use auxiliary modalities to increase feature diversity and design an enhanced prototype network for HSI classification.

However, these methods simply use the concept of prototype learning, usually based on the assumption that each sample contributes equally to the task. But in some recognition tasks, such as face recognition, some highly representative facial features (key areas such as eyes and nose) may be crucial to prototype learning of the category. If these key samples are simply regarded as equally important as other samples, it may lead to a decline in recognition performance.

B. Contrastive Learning

Contrastive learning aims to exploit the differences between different samples to learn useful feature representations. Its main idea is to learn more discriminative feature representations by placing similar samples close to each other and separating dissimilar samples from each other. It is widely used in areas such as face recognition [34], [35], [36], image retrieval [37], [38], [39], and recommendation systems [40], [41], [42].

In multisource RS data classification tasks, contrastive learning can promote the development and application of remote sensing image classification technology by strengthening feature representation learning and enhancing data diversity. Zhu et al. [43] explore the task of few-shot multisource data joint classification using contrastive learning and learning-aligned multisource feature representations in a self-supervised manner. Xu et al. [44] use contrast loss to update the extracted feature vector and enhance the distinction between categories to improve classification performance.

C. Dual-Source Data Classification

The dual-source data fusion classification scheme overcomes the technical limitations of obtaining high-resolution remote sensing images for target recognition and has been widely studied. At present, the dual-source data fusion classification scheme is mainly feature-level fusion classification. To improve the classification results, various technologies have been proposed, such as [45], which combines different feature learning methods such as pixel-level spatial-domain features and frequency-domain features to effectively extract the advantageous features of multisource data. Ma et al. [46] first perform data-level fusion to reduce the difference between dual-source data. To map dual-source features to the same feature space, feature migration and sharing are performed. Zhu et al. [47] also extract multiscale features from the perspectives of frequency domain and spatial domain, making full use of the texture detail features of the original data.

However, these methods focus more on how to effectively fuse the information of PAN and MS images, retain their respective characteristics and make full use of their complementarity, while ignoring the inherent properties of remote sensing data that have large intraclass differences and small interclass differences.

III. Method

This part provides a detailed description of the proposed DPFDA-Net. Initially, we provide a brief review of the overall design of DPFDA-Net. Following that, we introduce the designed PFE module. Finally, an FAM based on prototype decoupling is presented.

A. DPFDA-Net

Fig. 2 describes the overall structure of the proposed DPFDA-Net, comprising the following main components.

1) Prototype Feature Embedding: We first use a three-layer convolutional neural network to extract features of PAN and MS image patches and get f_p and f_m, respectively. Then, the PFE module is used to cluster the, respectively, extracted features, and representative prototype features P_p and P_m are identified by clustering similar feature points together. This process can be defined as follows:

$$P_p, P_m = \mathrm{PFE}(f_p, f_m). \tag{2}$$

Among them, the prototype contrastive learning (PCL) loss is used to reduce the distance between features of samples of the same category and, at the same time, widen the distance between samples of different categories.

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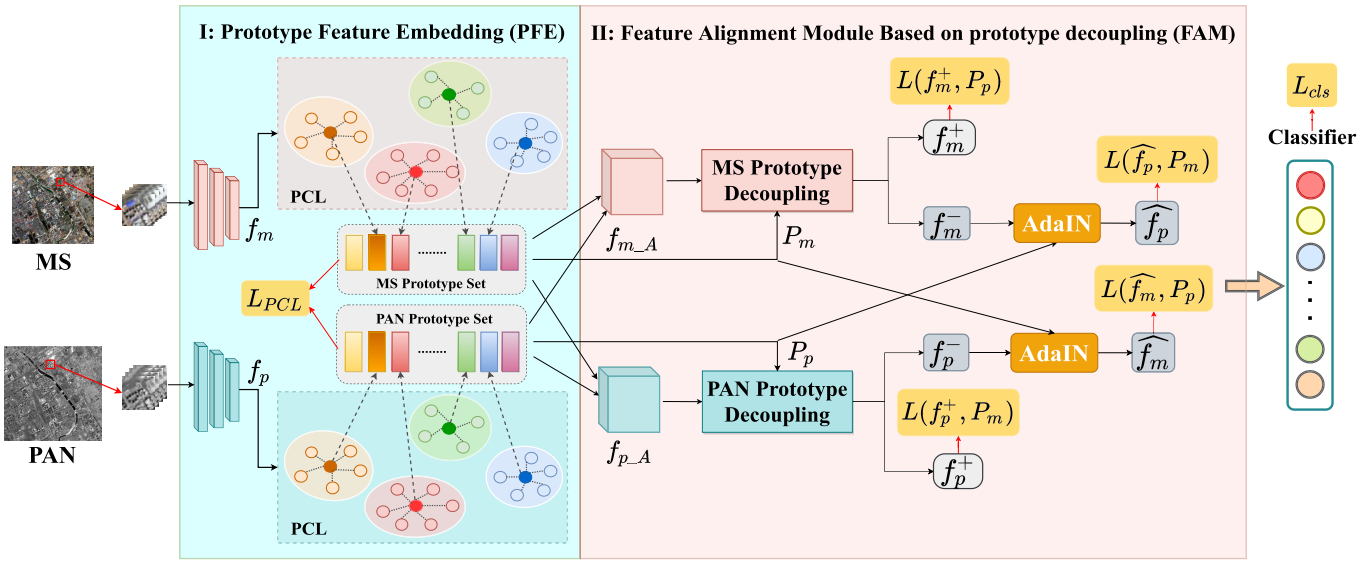


Fig. 2. Overall structure of our proposed DPFDA-Net. The dual-source image patches are first sent to the PFE module to learn prototype representation. Then, feature alignment is performed through FAM at the prototype feature level. Finally, the aligned dual-source features are fused and fed into the classifier for classification.

2) FAM Based on Prototype Decoupling: The PAN and MS prototype sets are used as inputs to the FAM based on prototype decoupling to obtain the aligned multisource features f_p' and f_m' as follows:

$$f_p', f_m' = \mathrm{FAM}(P_p, P_m). \tag{3}$$

3) Joint Training for Classification: The aligned multisource features are cascaded and fused for classification prediction of the network as follows:

$$f^{\mathrm{fusion}} = \mathrm{Concat}\left[P_{\mathrm{avg}}\left(f_p'\right), P_{\mathrm{avg}}\left(f_m'\right)\right] \tag{4}$$

where $P_{\mathrm{avg}}(\cdot)$ refers to global average pooling.

Subsequently, the network's overall training loss is designed as follows:

$$\begin{aligned} L_t = {} & L_{\mathrm{cls}} + L_{\mathrm{PCL}} + \alpha \cdot \left(L\left(f_p^+, P_m\right) + L\left(f_m^+, P_p\right)\right) \\ & + \beta \cdot \left(L\left(\widehat{f_m}, P_p\right) + L\left(\widehat{f_p}, P_m\right)\right) \end{aligned} \tag{5}$$

where L_t is the total loss. L_{cls} represents cross-entropy classification loss. L_{PCL} is the PCL loss. f_p^+ is the MS-related information obtained by decoupling the PAN prototype feature. Similarly, f_m^+ is the PAN-related information obtained by decoupling the MS prototype feature. $\widehat{f_m}$ and $\widehat{f_p}$ are intermediate quantities generated by the feature decoupling process. $L(f_p^+, P_m)$ and $L(f_m^+, P_p)$ are used to supervise prototype decoupling. $L(\widehat{f_m}, P_p)$ and $L(\widehat{f_p}, P_m)$ are used to achieve feature conversion. They are introduced in the following content. α and β are loss balance parameters, and we analyze their values in the hyperparameter experiment.

B. PFE Module

The improvement of the joint classification task performance based on PAN and MS is limited by the challenges of sample differences within the ground object category and sample similarities between categories. Prototype learning can better capture the commonalities of samples within the category and the differences between samples between categories by learning the prototype representation of each category. Therefore, this article uses the specific intersample relationships of the dual sources to design a PFE module for learning discriminative PAN and MS class prototype features, respectively. The details are as follows:

1) Sample Score Acquisition: To fully use the complementary advantages of dual-source data, we capture the single-source specific intersample relationships separately to further generate sample scores.

Specifically, for PAN data, we first calculate the relationship between the features of the ith sample and the features of other samples of the same category (denoted as $r_p(s_i, s_j)$) as follows:

$$r_p(s_i, s_j) = R(f_p(s_i, :), f_p(s_j, :)), \quad s_j \in S_i \tag{6}$$

where $R(\cdot)$ represents the relationship function. To avoid a high-dimensional disaster, we use cosine similarity. S_i represents a set of samples of the same category as s_i.

Similarly, $r_m(s_i, s_j)$ of the MS data is calculated as follows:

$$r_m(s_i, s_j) = R(f_m(s_i, :), f_m(s_j, :)), \quad s_j \in S_i. \tag{7}$$

Afterward, to smooth the weight distribution and minimize the impact of abnormal samples, we use the inverse multiquadratic (IMQ) function to regenerate sample relationships r_p' and r_m' because it is smooth and less sensitive to abnormal samples

$$r_*'(s_i, s_j) = \frac{1}{\sqrt{r_*^2(s_i, s_j) + C^2}} \tag{8}$$

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where C is a positive real number parameter. * refers to p or m.

Next, we capture the intersample relational difference matrix of PAN and MS by r'_p and r'_m and sum each row as follows:

$$\begin{cases} D_p = \sum_{i=1}^{b} \left(r'_p - r'_m\right)_i \\ D_m = \sum_{i=1}^{b} \left(r'_m - r'_p\right)_i \end{cases} \tag{9}$$

where D_* represents the specific intersample relationship of PAN or MS between the ith sample and other samples of the same category in the same batch.

After row summation, each value in D_* can reflect the difference between samples in the single-source data. The smaller the value of the sample, the smaller the difference between the sample and other samples, and vice versa. We believe that the smaller the value, the closer the sample to the prototype representation of the category and the more important the sample. Therefore, we assign a weight to each sample based on its score. In this way, important samples can be given higher weights, thereby making the most of the key information in each sample.

2) Prototype Generation: We normalize D_* to calculate the feature weight w_i for sample s_i

$$w_i = \frac{\max(D_*) - D_i}{\sum_{j=1}^{b} D_j} \tag{10}$$

where b represents the number of samples in a batch. $\max(\cdot)$ means taking the maximum value.

Then the discriminative class prototype p_k is finally obtained by weighted summation, calculated as follows:

$$p_k = \sum_{i=1}^{b} w_i \cdot f_i^k \tag{11}$$

where k represents the kth category. b represents the number of samples in a batch. f_i represents the feature vector of the ith sample.

Afterward, the class acquired prototype p_k is used for constructing a set of prototypes $P = \{p_1, p_2, \ldots, p_k\}$. Based on the above operations, the prototype sets P_p of PAN and the prototype sets P_m of MS are obtained.

3) Prototype Contrastive Learning: Contrastive learning optimization can weaken the difference between positive sample pairs and enhance the difference between negative sample pairs by learning the similarity relationship between samples [48]. Based on this feature, we introduce similarity measurement through contrastive learning method, forcing the learned representation to distinguish different categories, while reducing the problem of inconsistent performance of sample features of the same category in different sources due to differences in feature expression of dual-source data, and maximizing the consistency and robustness of data representation from the two sources.

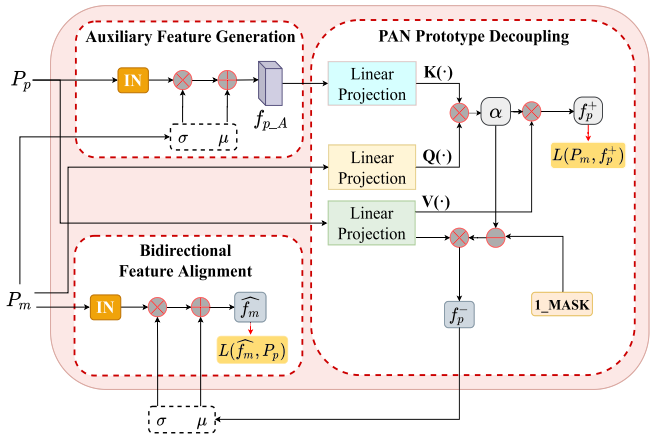


Fig. 3. Flowchart of FAM. Here, we take the decoupling of the PAN prototype feature as an example. IN represents the instance normalization operation that removes style features. μ and σ represent the mean and standard deviation, respectively.

Specifically, after obtaining the PAN and MS prototype sets, respectively, we perform PCL supervision as follows:

$$L_{\text{PCL}} = -\log \frac{\exp\left(R\left(P_p^i, P_m^i\right)\right)}{\sum_{n=1}^{b} \exp\left(R\left(P_p^i, P_m^n\right)\right)} \tag{12}$$

where P_p refers to the PAN prototype set, and P_m represents the MS prototype set. We define prototype features of the same category in P_p and P_m as positive sample pairs and prototype features of different categories as negative sample pairs. $R(\cdot)$ represents the cosine similarity measure the same as before.

PCL loss optimizes the distance between intra- and inter-class features in different embedding spaces. Combined with this loss, the network can learn more representative discriminative prototype features.

In general, the core idea of the PFE module is to obtain the discriminative prototype feature representation of the single source by capturing the specific intersample relationship of the single source, aggregating and encoding the sample features of the same category. The prototype feature can better represent the commonality of the entire category, thereby reducing the intraclass difference. At the same time, supervised clustering can independently learn the prototype of each category, and the categories will not interfere with each other, thereby increasing the interclass difference.

C. FAM Based on Prototype Decoupling

The class prototype vector obtained by prototype learning can be regarded as the center of a category in the feature space and has a more representative and robust feature representation. To construct an aligned multisource feature space and reduce feature expression differences, we decouple single-source features at the level of representative class prototypes and achieve feature alignment by supplementing the missing information of the single source through supervision. As shown in Fig. 3, here we take the decoupling of the PAN prototype feature as an example and the MS prototype is processed similarly.

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1) Auxiliary Feature Generation: To achieve single-source feature decoupling, considering that the AdaIN operation can adapt to different input feature distributions and style requirements, we first apply the AdaIN operation to generate PAN's auxiliary features f_{P_A} as follows:

$$f_{P_A} = \text{AdaIN}(P_p) = \sigma(P_m)\left(\frac{P_p - \mu(P_p)}{\sigma(P_p)}\right) + \mu(P_m) \tag{13}$$

where $\mu(\cdot)$ stands for the mean and $\sigma(\cdot)$ stands for the standard deviation. The AdaIN operation can perform style transfer, adjust PAN features into the style of MS features, and strengthen the internal consistency of matched dual-source features, thereby alleviating feature differences.

2) Prototype Decoupling: Since PAN and MS data describe the same ground scene, it is generally believed that the relevant features of PAN and MS are the spatial structure information, and the irrelevant features are the color style information. The PAN's auxiliary features f_{P_A} introduce the style of MS but maintain the same spatial structure as PAN. Therefore, it is feasible to decouple PAN features into MS-related features and MS-irrelevant features through auxiliary feature f_{P_A}.

As shown in Fig. 3, we improve the self-attention mechanism to achieve PAN prototype decoupling. The process is defined as follows:

$$\begin{cases} \alpha = \dfrac{e^{Q(P_m)\cdot K(f_{P_A})^T}}{\sum e^{Q(P_m)\cdot K(f_{P_A})^T}} \\ f_p^+ = \alpha \cdot V(P_p) \\ f_p^- = (1-\alpha)\cdot V(P_p) \end{cases} \tag{14}$$

where $Q(\cdot)$, $K(\cdot)$, and $V(\cdot)$ are the linear projection implemented through the fully connected layer. f_p^+ and f_p^- represent MS-related features and MS-irrelevant features obtained by decoupling the PAN prototype, respectively.

In (14), we define the weight α to represent the spatial structural relationship between the MS prototype representation P_m and the auxiliary features of PAN f_{P_A}. It uses MS prototype features P_m as the query to search for similar spatial structure information in the auxiliary features of PAN f_{P_A} (used as the key), thereby guiding PAN prototype features (used as the value) decomposition to obtain MS-related feature f_p^+, and further obtain MS-irrelevant features f_p^-.

At the same time, to supervise the decoupling of PAN prototype features to obtain MS-related features f_p^+, considering that $L1$ loss is easy to optimize and the gradient is more stable, we use it to reduce the distance between MS-related features f_p^+ and MS prototype P_m as follows:

$$L(f_p^+, P_m) = \frac{1}{k}\sum_{i=1}^{k} |f_{p_i}^+ - P_m^i| \tag{15}$$

where k represents the number of categories. $f_{p_i}^+$ refers to the ith category of MS-related features in PAN. P_m^i refers to the ith category prototype features of MS. Similarly, we optimize $L(f_m^+, P_p)$ to supervise the decomposition of MS prototype features and obtain PAN-related features f_m^+.

In general, the starting point of our prototype decoupling design is to decouple the data from source A into features related to source B data and features that are not related to source B data (assuming that the dual-source data are sources A and B, respectively, and here we take the decoupling of source A as an example), so as to further supplement the missing information of source A in a supervised learning manner, thereby achieving feature alignment of dual-source data.

3) Bidirectional Feature Alignment: To achieve feature alignment, based on f_p^- obtained by decoupling the PAN prototype features, we supplement the missing information of the single source in a supervised manner and establish information symmetry between dual-source features. Specifically, we establish the optimization of f_p^- by performing the feature conversion of MS→PAN. This process can be described as follows:

$$\widehat{f_m} = \text{AdaIN}(P_m) = \sigma(f_p^-)\left(\frac{P_m - \mu(P_m)}{\sigma(P_m)}\right) + \mu(f_p^-) \tag{16}$$

$$L\left(\widehat{f_m}, P_p\right) = \frac{1}{k}\sum_{i=1}^{k} |\widehat{f_{m_i}} - P_p^i| \tag{17}$$

where k represents the number of categories. P_p^i refers to the ith category prototype features of PAN.

We first perform the AdaIN operation on the MS prototype P_m to introduce the style of f_p^- and then get $\widehat{f_m}$. $\widehat{f_m}$ introduces the style of MS-irrelevant features but maintains similar structural information to PAN features. Then, further optimizing the distance between it and the PAN prototype P_p can establish the symmetric information flow of MS→PAN while supervising the decomposition of f_p^-. Similarly, for MS prototype feature decoupling, we optimize $L(\widehat{f_p}, P_m)$ to establish the symmetric information flow of PAN→MS. In this way, we supplement the missing information of the two sources in a bidirectional manner through supervision, which is conducive to achieving feature alignment and reducing the feature expression gap.

In general, we achieve feature alignment at the prototype level, which is more accurate and representative than alignment on common feature embedding which refers to directly performing feature mapping on the original input data through CNN without any other processing.

IV. Experiments

In this section, we provide a comprehensive introduction to dual-source RS datasets. Subsequently, we outline our experimental configuration and carry out a set of experiments to validate the enhanced performance achieved by our proposed design.

A. Data Introduction

The following three RS datasets were purchased and annotated by our research group. The detailed annotation of the data is shown in Fig. 4. These data were collected by the Gaofen-2 satellite. Their specific information is as follows.

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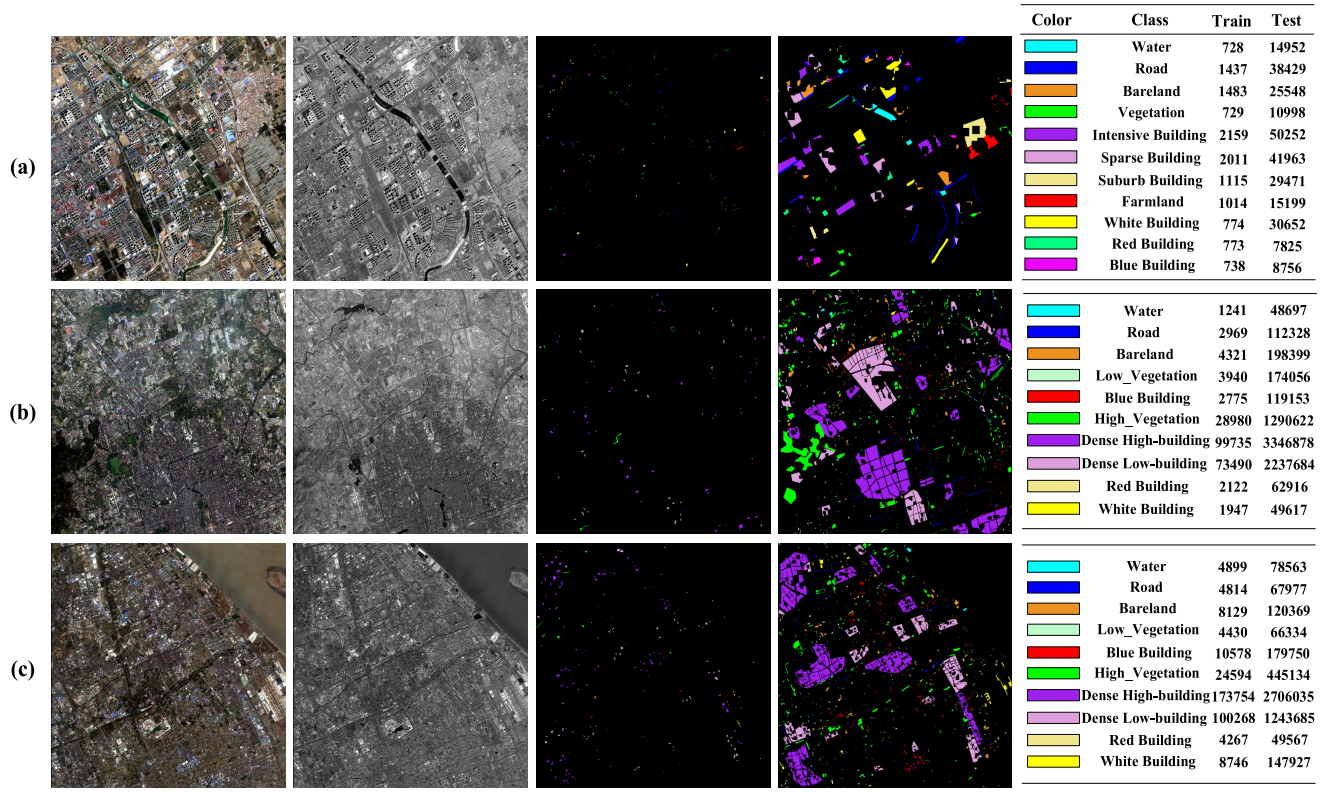


Fig. 4. (a) Distribution of the Hohhot area. (b) Distribution of the Beijing area. (c) Distribution of the Shanghai area. From left to right are the MS images, the PAN images, training sets, and test sets. On the right is detailed labeling information for each category.

1) Hohhot: The distribution of ground objects in Hohhot city is shown in Fig. 4(a). The MS data contain four channels: R, G, B, and NIR, and the image is in color. Its dimensions are 2001 × 2101 × 4 pixels, with a spatial resolution of 4 m. In contrast, the PAN data have only one channel and are, therefore, gray. It consists of 8004 × 8404 × 1 pixels, and the spatial resolution is 1 m. Notably, 11 distinct ground objects are identifiable, including water, road, bare land, vegetation, farmland, and six categories of building areas.

2) Beijing: Fig. 4(b) presents the distribution of ground objects in Beijing city. The MS data comprise 6908 × 7300 × 4 pixels with a spatial resolution of 4 m. The PAN data consist of 27 632 × 29 200 × 1 pixels with a spatial resolution of 1 m. We can identify that there are ten types of labeled classes, including water, road, bare land, two types of vegetation, and five different kinds of building areas.

3) Shanghai: As depicted in Fig. 4(c), the MS data depict a color image encompassing 6908 × 7300 × 4 pixels, with a spatial resolution of 4 m. In contrast, the PAN data represent a grayscale image with 27 632 × 29 200 × 1 pixels, and a spatial resolution of 1 m. It has ten kinds of labeled categories: water, road, bare land, two kinds of vegetation, and five different types of buildings.

The experiments conducted in this article use the workstation equipped with an RTX2080Ti 11-GB GPU and 128-GB RAM operating on the Ubuntu 16.04 LTS operating system. The training of the designed network is performed using the PyTorch framework.

B. Experimental Setting

We use three indicators, OA, AA, and Kappa [22], [49] coefficient, to quantitatively assess the effectiveness of the proposed approach. In the data preprocessing stage, we crop the PAN and MS images to 64 × 64 × 1 and 16 × 16 × 4, respectively. To ensure the fairness of the previous random sampling method, we conducted validation experiments based on completely separated training and test sets. The initial learning rate is set to 0.001 and we use the Adam optimizer to train our model. As shown in Fig. 5, considering the balance between model performance and efficiency, the network is trained for 30 epochs with a batch size of 128.

C. Selection of Hyperparameters

In this part, we analyze in detail the impact of different values of the two loss weight factors α and β in (5) on network training. The hyperparameter α is used to control the weight of the supervised prototype decoupling loss in the total loss, and the hyperparameter β is used to control the weight of the feature conversion loss in the total loss. This study uses the univariate method to conduct experiments and uses the OA value as the indicator to select and analyze the optimal value of the hyperparameter.

First, we fix the hyperparameter β to 0.5 to explore the optimal value of the hyperparameter α. The value range of α is set to 0.1–1.0, with a step size of 0.1. Next, we use the best value of α, start from 0.1, and gradually adjust β in steps

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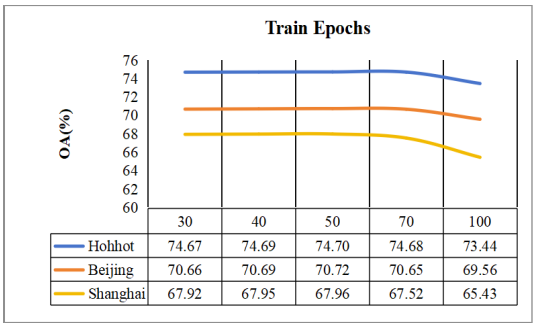


Fig. 5. Evaluation of training epochs. Experimental results show that when the number of training epochs increases, the model performance improves very little. When training for 100 epochs, the performance dropped significantly, and overfitting may have occurred. Furthermore, excessive training epochs consume a lot of computing resources and time but do not significantly improve the classification performance.

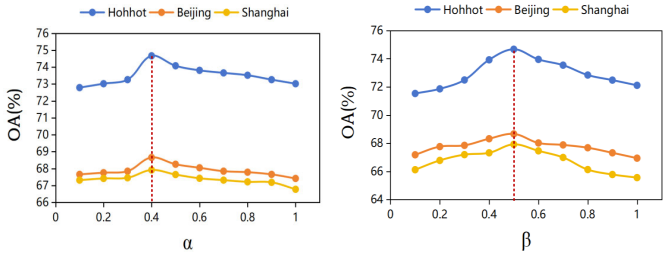


Fig. 6. Experimental OA values of hyperparameters with different values.

of 0.1, and record the experimental results as shown in Fig. 6. We find that on the three datasets, the best value of α is 0.4, while the best value of β is 0.5. In addition, the changing trend of overall accuracy (OA) is basically consistent across all the datasets. When the hyperparameter value deviates from the optimal value, the value of OA shows a downward trend.

These results show that in our experiments, the optimal values of α and β have a significant impact on the performance of the model. It is worth noting that these optimal values are consistent across multiple datasets, indicating that our method has a certain degree of universality. When α is far from 0.4, the OA value decreases. This may be because the feature decoupling is incomplete or excessive, resulting in redundant information being transferred between the two sources, thus affecting the classification performance. When β is far from 0.5, the OA value generally decreases. This is also because when achieving feature alignment, mutual conversion of dual-source features introduces repeated information.

D. Module Comparison

In this section of the study, we select Hohhot City as the experimental object and conduct a series of module comparison experiments. In particular, we improved the prototype generation method in the traditional prototype network in the PFE module and compared it with the traditional prototype network (TPN) [27]. In addition, we also compare the performance of the FAM method based on the AdaIN operation and the cross-attention (CA) mechanism with the basic CA mechanism. We integrate these components into the baseline network separately to avoid the influence of other modules. The experimental results are shown in Table I.

TABLE I
MODULE COMPARISON EXPERIMENT RESULTS ON HOHHOT AREA

Classes	TPN [27]	PFE	CA	FAM
c_1:Water	87.53	**89.62**	86.03	**90.28**
c_2:Road	**80.74**	79.61	**81.26**	78.61
c_3:Bareland	89.27	**90.26**	**88.05**	87.95
c_4:Vegetation	**87.47**	83.83	78.95	**82.79**
c_5:Intensive Building	60.73	**64.26**	63.61	**65.00**
c_6:Sparse Building	65.57	**65.93**	**69.38**	68.35
c_7:Suburb Building	77.74	**78.51**	**77.93**	76.98
c_8:Farmland	**91.05**	90.57	87.75	**96.99**
c_9:White Building	45.23	**56.20**	48.28	**48.96**
c_{10}:Red Building	**77.48**	74.40	62.95	**80.26**
c_{11}:Blue Building	72.56	**73.01**	71.72	**72.93**
OA(%)	72.11	**73.92**	72.49	**73.61**
AA(%)	75.94	**76.93**	74.17	**77.19**
Kappa(%)	68.55	**70.52**	68.82	**70.23**

1) PFE Versus TPN: As shown in Table I, the classification performance of our designed PFE is better than that of TPN. This is because the prototype in the traditional prototype network is obtained by averaging the category features, ignoring the differences between samples. The presence of abnormal samples will cause the learned prototype to shift in the embedding space, which is not conducive to classification. PFE collects the differences between samples and the importance of each sample and uses this to assign sample weights to obtain a more accurate prototype. At the same time, we introduce contrastive learning loss between the dual-source prototypes to further alleviate the problem of interclass differences. Therefore, PFE achieves higher accuracy than TPN.

2) FAM Versus CA: The experimental results show that our designed FAM module outperforms the traditional CA mechanism. The traditional CA mechanism mainly focuses on the fusion between the dual-branch features but ignores the mutual migration of the different information between the dual-source features, which often leads to the emergence of overmigration and redundancy. To solve this problem, we introduce the AdaIN operation to generate auxiliary features, which promotes the collaborative classification of dual-source data by adaptively migrating the advantage difference information to each other. This strategy enables our FAM module to better use the differences in the advantage features between different source data, thereby achieving higher classification accuracy.

E. Comparison to Other Methods

In this section, we have verified the performance improvement of the method designed in this article compared with other methods in recent years through qualitative and quantitative comparisons, including MFT: multimodal fusion transformer [50], AM^{3}Net: adaptive mutual-learning-based multimodal data fusion network [51], NNCNet: nearest-neighbor-based contrastive learning network [52], GCFnet: global collaborative fusion network [53], and ISSP-Net: interactive spatial–spectral perception network [45]. The parameter settings of the comparative experiments remain consistent with the original paper.

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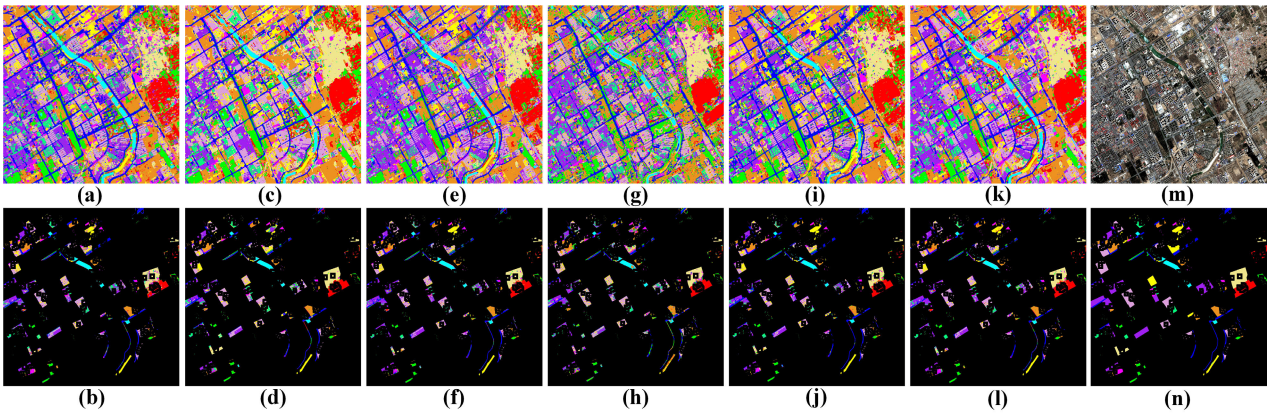


Fig. 7. Classification coloring maps of different methods on the Hohhot dataset. The first row is the overall colored image, and the second row is the coloring result of the labeled samples. (a) and (b) MFT. (c) and (d) AM^{3}Net. (e) and (f) NNCNet. (g) and (h) GCFNet. (i) and (j) ISSP-Net. (k) and (l) DPFDA-Net. (m) and (n) Ground truth.

TABLE II
QUANTITATIVE RESULTS ON HOHHOT AREA

Classes	MFT	AM^{3}Net	NNCNet	GCFnet	ISSP-Net	OURS
c_1	**96.18**	92.02	87.55	86.70	92.45	91.28
c_2	76.46	76.39	75.40	62.20	**82.82**	81.90
c_3	81.99	84.44	84.76	68.84	84.01	**87.26**
c_4	69.03	84.38	**85.24**	76.07	79.24	84.97
c_5	54.23	56.88	**67.63**	58.99	63.79	66.06
c_6	60.83	65.17	60.41	61.91	64.18	**68.06**
c_7	67.83	**86.66**	73.94	70.10	71.92	76.70
c_8	91.36	92.59	91.94	89.51	**95.12**	94.86
c_9	58.79	55.63	52.93	54.46	**62.65**	54.96
c_{10}	**82.46**	74.00	57.38	32.37	72.32	74.04
c_{11}	75.04	67.77	59.86	32.03	**78.13**	73.64
OA(%)	69.33	72.31	70.85	63.77	73.77	**74.67**
AA(%)	74.02	75.91	72.46	63.01	76.97	**77.61**
Kappa(%)	65.47	68.95	67.00	58.99	70.39	**71.38**
Time(s)	**15.62**	81.62	83.42	265.81	145.18	307.19

1) Comparison on Hohhot Area: Table II and Fig. 7 exhibit the experimental effect of our approach compared with other methods on the Hohhot dataset.

MFT adopts the transformer model that is superior to CNN and uses the concept of token to learn unique feature representations. Therefore, it achieves good performance in the recognition of c_1: water and c_{10}: red building.

Compared with MFT, AM^{3}Net shows improved performance in OA, AA, and Kappa indicators. In particular, the highest accuracy is achieved on c_7: suburb building. The building features of this category are quite different from those of other building categories, and the distribution is continuous and tight. Moreover, the color information of this category is rich. AM^{3}Net focuses on the specificity of each spectral channel of the HSI image and synergistically transmits spectral and spatial features between dual sources, achieving excellent classification performance.

The NNCNet method focuses on the semantic similarity features of neighborhood information and aligns multisource semantic information by building a contrastive learning framework. This method achieves the highest recognition accuracy in the two categories c_4: vegetation and c_5: dense building.

However, compared with the above three models, GCFnet performs poorly. This can be explained by the fact that this method only focuses on the collaborative fusion between the two branches and does not take into account the inherent semantic differences of dual-source RS data. Therefore, this method has certain limitations in performance and cannot fully use the complementary potential of dual-source data.

ISSP-Net fully exploits the advantages of dual-source data from the perspective of spatial and frequency domains and achieves competitive performance in categories such as c_2: road, c_8: farmland, c_9: white building, and c_{11}: blue building. This shows that focusing on spatial and spectral information can steadily improve the classification accuracy of certain categories.

It is noteworthy that our designed method outperforms the previously proposed methods in three indicators, which shows that the method is effective overall. In particular, it achieves the highest recognition accuracy on the categories of c_3: bare land and c_6: sparse building. The characteristics of these categories are relatively easy to capture by our designed PFE module by generating class prototypes. Although the performance of our method is not as good as the NNCNet on the two categories of c_4: vegetation and c_5: dense building, our method is not much different from it in accuracy. However, the accuracy of other categories is slightly lower than that of other methods because our model pays less attention to the extraction of spatial and spectral information from remote sensing data.

2) Comparison on Beijing Area: Table III and Fig. 8 exhibit the experimental results of the Beijing dataset. For the Beijing dataset, our method achieves higher OA, AA, and Kappa values. The contextual semantic information of remote sensing images is crucial for classification tasks, and NNCNet makes full use of this, slightly lower than our method in OA indicators. The classification performance of ISSP-Net on most categories shows a large difference from our method, showing a complementary trend. By comparison, it can be seen that ISSP-Net has weaker learning ability for categories such as c_7: dense high-building, c_8: dense low-building, and c_{10}: white building, which have weak spatial information, while the height information is rich and complex. The feature differences between the various building categories in the Beijing data are small, but our model achieves higher accuracy on these categories c_5: blue building, c_8: dense low-building, and c_{10}:

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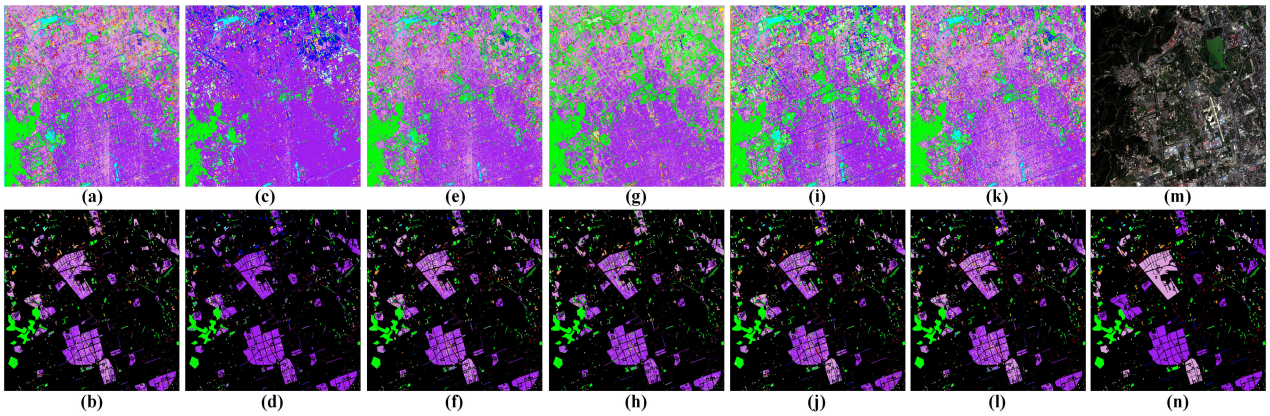


Fig. 8. Classification coloring maps of different methods on the Beijing dataset. The first row is the overall colored image, and the second row is the coloring result of the labeled samples. (a) and (b) MFT. (c) and (d) AM^{3}Net. (e) and (f) NNCNet. (g) and (h) GCFNet. (i) and (j) ISSP-Net. (k) and (l) DPFDA-Net. (m) and (n) Ground truth.

TABLE III
QUANTITATIVE RESULTS ON BEIJING AREA

Classes	MFT	AM^{3}Net	NNCNet	GCFnet	ISSP-Net	OURS
c_1	62.39	61.37	64.89	2.09	**75.12**	63.77
c_2	9.62	16.66	14.59	0.51	**26.21**	15.88
c_3	53.50	29.91	64.58	54.49	**82.07**	70.44
c_4	71.37	78.46	72.32	51.73	**92.11**	79.66
c_5	71.83	50.09	66.52	45.01	70.4	**74.67**
c_6	92.30	72.95	90.89	90.75	**95.36**	91.47
c_7	54.35	**91.57**	69.83	66.28	66.53	87.07
c_8	69.03	13.09	56.97	44.72	52.88	**70.42**
c_9	57.64	43.24	67.97	44.30	**78.55**	74.51
c_{10}	76.73	61.95	79.27	50.00	67.17	**86.57**
OA(%)	65.27	61.01	68.69	61.47	68.01	**70.66**
AA(%)	61.88	51.93	64.78	44.99	70.64	**71.45**
Kappa(%)	51.27	40.03	55.09	44.15	55.05	**59.51**
Time(s)	**536.3**	4880.98	2741.95	7736.46	3471.25	5302.62

TABLE IV
QUANTITATIVE RESULTS ON SHANGHAI AREA

Classes	MFT	AM^{3}Net	NNCNet	GCFnet	ISSP-Net	OURS
c_1	73.37	28.93	73.74	71.34	**79.96**	78.46
c_2	17.57	5.38	42.24	41.16	**49.34**	36.08
c_3	73.71	6.82	81.86	83.16	83.07	**84.46**
c_4	61.44	24.62	80.23	69.92	**82.6**	80.61
c_5	70.15	52.60	75.75	83.82	**88.98**	83.20
c_6	79.09	82.36	80.85	82.26	**85.09**	83.76
c_7	73.83	**94.09**	70.32	64.22	68.4	70.82
c_8	39.75	10.84	47.70	45.87	49.02	**49.34**
c_9	53.42	35.81	36.40	10.20	45.11	**56.47**
c_{10}	92.77	83.05	96.15	93.49	95.9	**98.71**
OA(%)	65.29	65.29	66.42	62.66	66.88	**67.92**
AA(%)	63.51	42.45	68.52	64.54	**72.74**	72.19
Kappa(%)	46.37	38.23	49.37	43.35	51.33	**52.04**
Time(s)	**308.97**	1712.7	1723.11	2727.74	1312.72	2082.14

white building. This fully demonstrates that our PFE module has learned different building category prototypes, thereby effectively distinguishing these categories.

3) Comparison on Shanghai Area: As presented in Table IV and Fig. 9, for the Shanghai dataset, our designed method achieves the highest accuracy on most categories including c_3: bareland, c_8: dense low-building, c_9: red building, and c_{10}: white building. In particular, the differences in the features of c_8, c_9, and c_{10}, the several categories of objects, are very small, which shows that our designed model perceives the high difference between the three categories and effectively distinguishes them. The effective alignment of the feature space prevents the learning of the prototype features of these categories from having a large deviation, thus achieving superior OA, AA, and Kappa values. AM^{3}Net fully exploits the spatial information of dual-source data and achieves the highest accuracy on c_7: dense high-building. In addition, ISSP-Net has slightly higher classification performance than our model on some categories due to its spatial–spectral feature extraction capability.

F. Ablation Study

This section aims to explore the impact of each component on the model performance and conducts ablation studies to evaluate their effects. We conduct ablation experiments under the same experimental settings. As shown in Table V, OA, AA, and Kappa values are all improved by more than 2% when the baseline model is combined with the PFE component. This is because the prototype features extracted by PFE enlarge the feature differences between different categories while reducing the distance between features of the same category, effectively solving the problem of "intraclass and interclass differences." In addition, the accuracy of the three indicators is improved by nearly 2% when the baseline model is combined with the FAM component. This is because FAM helps find a unified multisource feature space by aligning the features extracted from the two sources, thereby improving the classification performance.

TABLE V
ABLATION STUDY RESULTS OF EACH COMPONENT ON HOHHOT AREA

Baseline	PFE	FAM	OA(%)	AA(%)	Kappa(%)
✓			71.86	74.71	68.17
✓	✓		73.92	76.93	70.52
✓		✓	73.61	77.19	70.23
✓	✓	✓	74.67	77.61	71.38

It is worth noting that the best classification accuracy is achieved by combining the baseline model with both PFE and FAM components. FAM performs feature alignment at the

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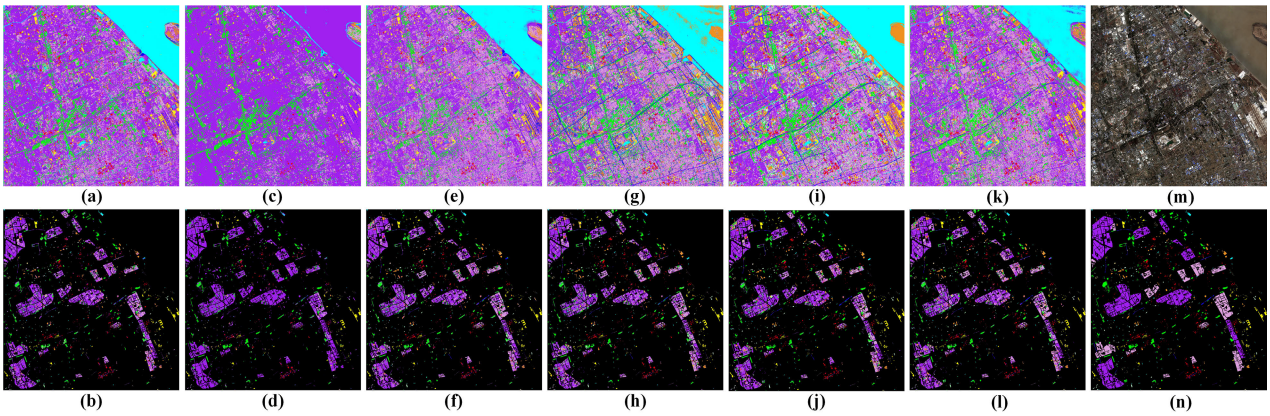


Fig. 9. Classification coloring maps of different methods on the Shanghai dataset. The first row is the overall colored image, and the second row is the coloring result of the labeled samples. (a) and (b) MFT. (c) and (d) AM^{3}Net. (e) and (f) NNCNet. (g) and (h) GCFNet. (i) and (j) ISSP-Net. (k) and (l) DPFDA-Net. (m) and (n) Ground Truth.

prototype feature level, which is more accurate than alignment on common features, thus achieving performance comparable to the SOTA methods.

The above results show that each component has a significant impact on the improvement of model performance. The introduction of PFE and FAM improves the performance of the model through feature engineering and feature alignment, respectively, which confirms the effectiveness of these components in solving multisource data classification problems.

V. Conclusion

This research designs a DPFDA-Net, which aims to address the challenges of category differences and inconsistent feature embedding space of dual-source RS data. Through a series of experiments, the two components we designed are verified to effectively enhance the classification effect. First, the PFE module introduces a prototype learning method to learn representative category prototype features in dual-source data and supervise them through contrastive learning. This effectively solves the problem of large intraclass differences and small interclass differences and improves the accuracy of classification. Second, the FAM module aligns dual-source category features at the prototype level, which is more targeted and helps find a consistent dual-source feature space. This feature alignment improves the robustness of the classification model. Although the method designed in this article achieves satisfactory performance, it also has some limitations. First, the computational complexity is large, resulting in a long test time. To further improve practicality, future research can explore the design of a lighter network structure to achieve more efficient calculations. In addition, the algorithm details can be further optimized to improve the efficiency and scalability of the method.

References

[1] J. Zhao et al., "OrientedFormer: An end-to-end transformer-based oriented object detector in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5640816.

[2] T. Feng, L. Jiao, H. Zhu, and L. Sun, "A novel object re-track framework for 3D point clouds," in *Proc. 28th ACM Int. Conf. Multimedia*, Oct. 2020, pp. 3118–3126.

[3] Y. Han et al., "SSMU-net: A style separation and mode unification network for multimodal remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5407115.

[4] J. Tan et al., "Revisiting spatial-frequency information integration from a hierarchical perspective for panchromatic and multi-spectral image fusion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 25922–25931.

[5] Q. Xu, Y. Li, J. Nie, Q. Liu, and M. Guo, "UPanGAN: Unsupervised pansharpening based on the spectral and spatial loss constrained generative adversarial network," *Inf. Fusion*, vol. 91, pp. 31–46, Mar. 2023.

[6] Z. Li, J. Li, F. Zhang, and L. Fan, "CADUI: Cross-attention-based depth unfolding iteration network for pansharpening remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5402420.

[7] M. Zhou, J. Huang, D. Hong, F. Zhao, C. Li, and J. Chanussot, "Rethinking pan-sharpening in closed-loop regularization," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 10, pp. 1–15, Oct. 2024.

[8] X. Li et al., "A collaborative learning tracking network for remote sensing videos," *IEEE Trans. Cybern.*, vol. 53, no. 3, pp. 1954–1967, Mar. 2023.

[9] M. Ma et al., "MBSI-net: Multimodal balanced self-learning interaction network for image classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 5, pp. 3819–3833, May 2024.

[10] M. Ma et al., "A multimodal hyper-fusion transformer for remote sensing image classification," *Inf. Fusion*, vol. 96, pp. 66–79, Aug. 2023.

[11] M. Ma et al., "Transfer representation learning meets multimodal fusion classification for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5632415.

[12] W. Ma et al., "A collaborative correlation-matching network for multimodality remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5409718.

[13] W. Li et al., "An effective multimodel fusion method for SAR and optical remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 5881–5892, 2023.

[14] K. Zhang et al., "Panchromatic and multispectral image fusion for remote sensing and Earth observation: Concepts, taxonomy, literature review, evaluation methodologies and challenges ahead," *Inf. Fusion*, vol. 93, pp. 227–242, May 2023.

[15] W. Ye, B. Lin, J. Lao, Y. Liu, and Z. Lin, "MRA-IDN: A lightweight super-resolution framework of remote sensing images based on multi-scale residual attention fusion mechanism," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 7781–7800, 2024.

[16] S. Teruggi, E. Grilli, M. Russo, F. Fassi, and F. Remondino, "A hierarchical machine learning approach for multi-level and multi-resolution 3D point cloud classification," *Remote Sens.*, vol. 12, no. 16, p. 2598, Aug. 2020.

[17] G. Xie, R. Nie, J. Cao, H. Li, and J. Li, "A deep multiresolution representation framework for pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5517216.

[18] Y. Liao et al., "A two-stage mutual fusion network for multispectral and panchromatic image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5413218.

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[19] P. Wang, H. Yao, C. Li, G. Zhang, and H. Leung, "Multiresolution analysis based on dual-scale regression for pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5406319.
[20] J. Zhang et al., "Pan-sharpening with wavelet-enhanced high-frequency information," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5402914.
[21] L. Si et al., "A trusted generative-discriminative joint feature learning framework for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5601814.
[22] W. Ma et al., "Intra- and intersource interactive representation learning network for remote sensing images classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5401515.
[23] X. Zhao et al., "Fractional Fourier image transformer for multimodal remote sensing data classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 2, pp. 2314–2326, Feb. 2024.
[24] G. Cheng et al., "Prototype-CNN for few-shot object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5604610.
[25] C. Ding et al., "Integrating prototype learning with graph convolution network for effective active hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504816.
[26] C. Yu et al., "Distillation-constrained prototype representation network for hyperspectral image incremental classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5507414.
[27] G. Cheng et al., "SPNet: Siamese-prototype network for few-shot remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5608011.
[28] X. Gu, C. Zhang, Q. Shen, J. Han, P. P. Angelov, and P. M. Atkinson, "A self-training hierarchical prototype-based ensemble framework for remote sensing scene classification," *Inf. Fusion*, vol. 80, pp. 179–204, Apr. 2022.
[29] H. Wang, M. Zhang, W. Li, Y. Gao, Y. Gui, and Y. Zhang, "Unbalanced class learning network with scale-adaptive perception for complicated scene in remote sensing images segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4406712.
[30] Y. Zheng, S. Liu, H. Chen, and L. Bruzzone, "Hybrid FusionNet: A hybrid feature fusion framework for multisource high-resolution remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5401714.
[31] Y. Yang et al., "An explainable spatial–frequency multiscale transformer for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5907515.
[32] X. Cheng, F. Shi, X. Liu, M. Zhao, and S. Chen, "A novel deep class-imbalanced semisupervised model for wind turbine blade icing detection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 6, pp. 2558–2570, Jun. 2022.
[33] W. Wang, F. Liu, and L. Xiao, "I^2MEP-net: Inter- and intra-modality enhancing prototypical network for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516815.
[34] Y. Yang, W. Hu, H. Lin, and H. Hu, "Robust cross-domain pseudo-labeling and contrastive learning for unsupervised domain adaptation NIR-VIS face recognition," *IEEE Trans. Image Process.*, vol. 32, pp. 5231–5244, 2023.
[35] Y. Song and F. Wang, "CoReFace: Sample-guided contrastive regularization for deep face recognition," *Pattern Recognit.*, vol. 152, Aug. 2024, Art. no. 110483.
[36] X. Luan, Z. Ding, L. Liu, W. Li, and X. Gao, "A symmetrical Siamese network framework with contrastive learning for pose-robust face recognition," *IEEE Trans. Image Process.*, vol. 32, pp. 5652–5663, 2023.
[37] M. Huang, L. Dong, W. Dong, and G. Shi, "Supervised contrastive learning based on fusion of global and local features for remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5208513.
[38] H. Cao, L. Huang, J. Nie, and Z. Wei, "Unsupervised deep hashing with fine-grained similarity-preserving contrastive learning for image retrieval," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 5, pp. 4095–4108, May 2024.
[39] C. Hu, C. Zhang, and G. H. Lee, "Unsupervised feature representation learning for domain-generalized cross-domain image retrieval," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 11016–11025.
[40] F. Guo, Z. Wang, X. Wang, Q. Lu, and S. Ji, "Dual-view multi-modal contrastive learning for graph-based recommender systems," *Comput. Electr. Eng.*, vol. 116, May 2024, Art. no. 109213.
[41] J. Yu, X. Xia, T. Chen, L. Cui, N. Q. V. Hung, and H. Yin, "XSimGCL: Towards extremely simple graph contrastive learning for recommendation," *IEEE Trans. Knowl. Data Eng.*, vol. 36, no. 2, pp. 913–926, Feb. 2024.
[42] Y. Yu, Q. Liu, L. Wu, R. Yu, S. L. Yu, and Z. Zhang, "Untargeted attack against federated recommendation systems via poisonous item embeddings and the defense," in *Proc. AAAI Conf. Artif. Intell.*, Jun. 2023, vol. 37, no. 4, pp. 4854–4863.
[43] H. Zhu et al., "Few-shot MS and PAN joint classification with improved cross-source contrastive learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5406615.
[44] Y. Xu et al., "Attention-based contrastive learning for few-shot remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5620317.
[45] W. Ma, H. Zhang, M. Ma, C. Chen, and B. Hou, "ISSP-net: An interactive spatial–spectral perception network for multimodal classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4412014.
[46] W. Ma et al., "An adaptive migration collaborative network for multimodal image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 8, pp. 1–15, Aug. 2024.
[47] H. Zhu et al., "ConvGRU-based multiscale frequency fusion network for PAN-MS joint classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5406415.
[48] H. Liu, J. Feng, L. Kong, D. Tao, Y. Chen, and M. Zhang, "Graph contrastive learning meets graph meta learning: A unified method for few-shot node tasks," in *Proc. ACM Web Conf.*, May 2024, pp. 365–376.
[49] X. Zhang, W. Li, C. Gao, Y. Yang, and K. Chang, "Hyperspectral pathology image classification using dimension-driven multi-path attention residual network," *Expert Syst. Appl.*, vol. 230, Nov. 2023, Art. no. 120615.
[50] S. K. Roy, A. Deria, D. Hong, B. Rasti, A. Plaza, and J. Chanussot, "Multimodal fusion transformer for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5515620.
[51] J. Wang, J. Li, Y. Shi, J. Lai, and X. Tan, "AM^3net: Adaptive mutual-learning-based multimodal data fusion network," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 8, pp. 5411–5426, Aug. 2022.
[52] M. Wang, F. Gao, J. Dong, H.-C. Li, and Q. Du, "Nearest neighbor-based contrastive learning for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501816.
[53] H. Zhao et al., "GCFnet: Global collaborative fusion network for multispectral and panchromatic image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5632814.



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